CITS5508 Practice Exam — Chapter 9 — ANSWER KEY

Question 1.

(a) *(3 marks)* **Clustering** = the unsupervised task of grouping **similar instances** into clusters. In **classification** a label is the **true target class** (supplied during training); in **clustering** an instance's "label" is just the **index of the cluster** the algorithm assigned it to (discovered, not a ground-truth class).

(b) *(4 marks)* Any four: customer segmentation; data analysis; dimensionality reduction (cluster affinities); feature engineering; anomaly detection; semi-supervised learning; search engines (image similarity); image segmentation.

(c) *(5 marks)* **k-means loop:** (1) **assign** each instance to the nearest **centroid**; (2) **update** each centroid to the **mean** of its assigned instances. Repeat until centroids stop moving. It is **guaranteed to converge** because the mean squared distance to centroids **can only decrease** each step and is bounded below by 0. It can converge to a **suboptimal (local) optimum** depending on the **random centroid initialisation** (mitigated by `n_init`/k-means++).

Question 2.

(a) *(3 marks)* **Inertia** (sum of squared distances to the nearest centroid) **always decreases as k increases** — more centroids mean every instance is closer to one — so minimising inertia would just pick the largest k. It cannot, by itself, indicate the right number of clusters.

(b) *(4 marks)* - **Elbow method:** plot **inertia vs k** and pick the **elbow** — the k after which inertia drops only slowly (diminishing returns). Coarse but quick. - **Silhouette score:** the mean **silhouette coefficient** over all instances; plot it vs k and pick the **maximum**. More precise (but more expensive).

(c) *(3 marks)* `silhouette = (b − a) / max(a, b) = (2.0 − 0.5) / max(0.5, 2.0) = 1.5 / 2.0 =`**`0.75`**. A value close to **+1** means the instance is **well inside its own cluster and far from the nearest other cluster** (a good assignment). The coefficient ranges from **−1 to +1** (≈0 = on a boundary; <0 = likely in the wrong cluster).

(d) *(2 marks)* Any two: clusters with **different sizes**, **different densities**, or **non-spherical/elliptical shapes**; also if features are **not scaled**.

Question 3.

(a) *(5 marks)* **DBSCAN** defines clusters as **continuous regions of high density**. - **ε-neighbourhood:** the set of instances within distance **ε** of a given instance. - **Core instance:** an instance with at least `min_samples` instances in its ε-neighbourhood (i.e. located in a dense region). - Instances in a core instance's neighbourhood join its cluster; chains of neighbouring core instances form one cluster. - **Anomaly:** an instance that is **neither a core instance nor in the neighbourhood of one** (labelled −1).

(b) *(4 marks)* Advantages over k-means (any two): finds clusters of **arbitrary shape**; **does not require specifying k**; **robust to outliers** (labels them as anomalies). Limitation (one): struggles when cluster **densities vary** a lot; and/or **scales poorly** (≈ $O(m^2 \cdot n)$).

(c) *(3 marks)* Train a separate classifier (e.g. a **`KNeighborsClassifier`**) on the DBSCAN **core instances** and their cluster labels, then use it to predict the cluster of a new instance (optionally setting a maximum distance so far-away points are labelled anomalies).

Question 4.

(a) *(3 marks)* A **GMM** assumes the data was generated from a **mixture of k Gaussian distributions** with unknown parameters: each instance is drawn from one of the k Gaussians (chosen by cluster weight), and instances from one Gaussian form an (typically **ellipsoidal**) cluster with its own mean, covariance (shape/size/orientation), and weight.

(b) *(5 marks)* **EM steps:** (1) **Expectation:** estimate the **probability (responsibility)** that each instance belongs to each cluster, given current parameters; (2) **Maximization:** update each cluster's weight/mean/covariance using **all** instances, each weighted by its responsibility. Repeat to convergence. Differences from k-means: EM uses **soft assignments** (probabilities) rather than hard ones, and it learns not just the cluster **centres** but also their **size, shape, orientation, and weights** (covariance matrices).

(c) *(4 marks)* Use a **theoretical information criterion** — **BIC** and **AIC** (both penalise the number of parameters and reward fit; minimise them over k). A variant that auto-selects clusters: **`BayesianGaussianMixture`** (sets unnecessary clusters' weights to ≈0).

Question 5.

(a) *(4 marks)* **Anomaly detection** assumes the training set **may be contaminated** with outliers and aims to detect (and often remove) them — so the method must be **robust to outliers in training**. **Novelty detection** assumes the training set is **clean** (no outliers) and aims to flag **new** instances that differ from everything seen during training. The key difference is the **assumption about the training set's cleanliness**.

(b) *(4 marks)* A trained GMM gives a **density** (PDF / `score_samples`) at any point; instances in **low-density regions** are flagged as anomalies. Choose the **density threshold** to match the **expected anomaly rate** — e.g. if ~2% of products are defective, set the threshold at the **2nd percentile** of densities. Too many **false positives** → lower the threshold; too many **false negatives** → raise it. This tuning is exactly the **precision/recall trade-off**.

(c) *(4 marks)* Any two, e.g.: - **Isolation Forest** — random trees split the space randomly; anomalies get **isolated in fewer splits** (shorter average path) because they're far from other instances. - **Local Outlier Factor (LOF)** — compares an instance's **local density** to that of its neighbours; an anomaly is **much less dense** than its neighbours. - **One-class SVM** — separates the data from the origin in feature space; points outside the learned region are anomalies (good for novelty detection). - **PCA reconstruction error** — anomalies reconstruct **poorly** (large error).